1. Seaborn Basics & Setup

- **Import:** `import seaborn as sns`
- **Import Matplotlib:** `import matplotlib.pyplot as plt` (often used together)
- **Load Sample Data:** `df = sns.load_dataset('tips')`
- **Set Style:** `sns.set_style('whitegrid')`, `sns.set_palette('viridis')`

2. Distribution Plots

Histograms & KDEs

- **Histogram (`histplot`):** Displays univariate distribution.

```
sns.histplot(data=df, x='total_bill',
 kde=True, bins=20)
 plt.title('Distribution of Total Bill')
 plt.show()
```

- **Kernel Density Estimate (`kdeplot`):** Smoothed representation of distribution.

```
sns.kdeplot(data=df, x='total_bill',
 fill=True)
 plt.title('KDE of Total Bill')
 plt.show()
```

- **Displot (`displot`):** Combines hist, kde, rug.

```
sns.displot(data=df, x='total_bill',
 kind='hist', kde=True)
 plt.title('Displot of Total Bill')
 plt.show()
```

3. Relational Plots

Scatterplots & Lineplots

- **Scatter Plot (`scatterplot`):** Shows relationship between two numerical variables.

```
sns.scatterplot(data=df, x='total_bill',
 y='tip', hue='time', style='smoker',
 size='size')
 plt.title('Total Bill vs. Tip by Time and
 Smoker')
 plt.show()
```

  - `hue`: Color points by category.
  - `style`: Vary marker style by category.
  - `size`: Vary marker size by numerical variable.
- **Line Plot (`lineplot`):** Shows relationship, often with time series or ordered data.

```
fmri = sns.load_dataset('fmri')
 sns.lineplot(data=fmri, x='timepoint',
 y='signal', hue='event', style='region')
 plt.title('FMRI Signal Over Time')
 plt.show()
```

  - Often aggregates data by default (e.g., mean and confidence interval).
- **Relplot (`relplot`):** Figure-level function for scatter/line plots with faceting.

```
sns.relplot(data=df, x='total_bill', y='tip',
 col='time', row='smoker', kind='scatter')
 plt.suptitle('Tips by Total Bill, Time, and
 Smoker', y=1.02)
 plt.show()
```

  - `col`, `row`: Create separate subplots (facets) for categories.
  - `kind`: 'scatter' or 'line'.

4. Categorical Plots

Bar, Box, Violin, Swarm, Count

- **Bar Plot (`barplot`):** Shows mean of numerical variable for each category.

```
sns.barplot(data=df, x='day', y='total_bill',
 hue='sex', errorbar='sd')
 plt.title('Average Total Bill by Day and
 Sex')
 plt.show()
```

  - `errorbar`: Shows confidence interval (default) or standard deviation.
- **Count Plot (`countplot`):** Shows count of observations in each category.

```
sns.countplot(data=df, x='day', hue='time')
 plt.title('Number of Customers by Day and
 Time')
 plt.show()
```

- **Box Plot (`boxplot`):** Shows distribution using quartiles.

```
sns.boxplot(data=df, x='day', y='total_bill',
 hue='smoker')
 plt.title('Total Bill Distribution by Day
 and Smoker')
 plt.show()
```

  - Median, quartiles (25th, 75th percentile), and outliers.
- **Violin Plot (`violinplot`):** Combines box plot with KDE.

```
sns.violinplot(data=df, x='day',
 y='total_bill', hue='sex', split=True)
 plt.title('Total Bill Distribution by Day
 and Sex')
 plt.show()
```

  - `split=True`: Splits violins for hue categories.
- **Swarm Plot (`swarmplot`):** Shows all individual data points, avoids overlap.

```
sns.swarmplot(data=df, x='day',
 y='total_bill', hue='sex', dodge=True)
 plt.title('Total Bill by Day and Sex (All
 Points)')
 plt.show()
```

  - Often used with box/violin plots for more detail.
- **Catplot (`catplot`):** Figure-level function for categorical plots with faceting.

```
sns.catplot(data=df, x='day', y='total_bill',
 hue='sex', col='time', kind='box')
 plt.suptitle('Total Bill by Day, Sex, and
 Time', y=1.02)
 plt.show()
```

  - `kind`: 'bar', 'box', 'violin', 'swarm', 'point', 'strip', 'count'.

5. Matrix Plots

Heatmaps & Pairplots

- **Heatmap (`heatmap`):** Visualizes matrix data, like correlation matrices.

```
iris = sns.load_dataset('iris')
 corr_matrix = iris.drop('species',
 axis=1).corr()
 sns.heatmap(corr_matrix, annot=True,
 cmap='coolwarm', fmt='.2f')
 plt.title('Correlation Matrix of Iris
 Features')
 plt.show()
```

  - `annot=True`: Display correlation values on the heatmap.
  - `cmap`: Color map (e.g., 'viridis', 'coolwarm', 'RdBu').
  - **Use Cases:** Correlation matrices, feature importance, confusion matrices.
- **Pair Plot (`pairplot`):** Plots pairwise relationships in a dataset.

```
sns.pairplot(data=iris, hue='species',
 diag_kind='kde')
 plt.suptitle('Pairwise Relationships of Iris
 Features', y=1.02)
 plt.show()
```

  - Diagonal plots show univariate distribution (hist, kde).
  - Off-diagonal plots show bivariate relationships (scatter).
  - `hue`: Color points by category.
  - **Use Cases:** Exploring multivariate relationships, initial data exploration.

6. Regression Plots

- **Regplot (`regplot`):** Plots data and a linear regression model fit.

```
sns.regplot(data=df, x='total_bill', y='tip',
 scatter_kws={'alpha':0.3})
 plt.title('Total Bill vs. Tip with
 Regression Line')
 plt.show()
```

- **Lmplot (`lmplot`):** Figure-level function for regression plots with faceting.

```
sns.lmplot(data=df, x='total_bill', y='tip',
 hue='smoker', col='time')
 plt.suptitle('Regression of Tip vs. Total
 Bill by Smoker and Time', y=1.02)
 plt.show()
```

7. Advanced Customization

- **Figure Size:** `plt.figure(figsize=(10, 6))`
- **Titles:** `plt.title('My Plot Title')`
- **Labels:** `plt.xlabel('X-axis Label')`, `plt.ylabel('Y-axis Label')`
- **Legends:** `plt.legend(title='Category')`
- **Saving Plots:** `plt.savefig('my_plot.png', dpi=300, bbox_inches='tight')`
- **Subplots:** `fig, axes = plt.subplots(1, 2, figsize=(12, 5))`
- **Style Context:** `sns.set_context('talk')` (paper, notebook, talk, poster)